SPINE · NEET-SS / FET / INI-SS

Spinal tumours and metastases

1. Overview

- **Metastases are by far the commonest spinal tumours** (about 95% of spinal tumours in adults over 40). The spine is the **commonest site of skeletal metastasis**; **thoracic spine commonest (about 70%)**, then lumbar, cervical.
- **Primary sources (adults): breast, prostate, lung** (together the majority), **kidney, thyroid** (both **hypervascular**: **pre-operative embolisation**), myeloma (haematological). "**BLT with a Kosher Pickle**" (breast, lung, thyroid, kidney, prostate).
- **Spread:** haematogenous (**Batson's plexus**), starting in the **vertebral body (posterior part)**, then **pedicle**; **pedicle destruction ("winking owl" sign)** on AP radiograph is an early radiographic sign. **The disc is usually spared** (vs infection).
- **Location of tumours in the spine:**
  - **Vertebral body:** metastases, myeloma, lymphoma, haemangioma, chordoma, giant cell tumour, Ewing, osteosarcoma.
  - **Posterior elements: osteoid osteoma, osteoblastoma, aneurysmal bone cyst**, osteochondroma.
  - **Intradural extramedullary: meningioma, schwannoma (neurofibroma)** (dumbbell lesions widen foramina).
  - **Intramedullary: ependymoma** (adults), **astrocytoma** (children), haemangioblastoma.

2. Primary benign lesions

Lesion	Typical patient / site	Key features	Treatment
Haemangioma	Commonest benign spinal lesion (about 10% at autopsy), thoracic	**Corduroy/vertical striations** on radiograph, **polka-dot** on axial CT; T1 and T2 bright (fat)	Usually incidental; aggressive symptomatic: embolisation, vertebroplasty, radiotherapy, surgery
Osteoid osteoma	Young males, **posterior elements, lumbar**	**Painful scoliosis** (lesion on the **concave side of the apex**), night pain relieved by **NSAIDs**; nidus under 1.5-2 cm on CT	NSAIDs; **CT-guided radiofrequency ablation** (caution near neural structures) or excision
Osteoblastoma	Young, **posterior elements**, cervical	Nidus over 2 cm, less NSAID-responsive, can be aggressive	**Marginal/en bloc excision** (recurrence)
Aneurysmal bone cyst	Under 20, posterior elements extending into the body	**Fluid-fluid levels** on MRI, expansile	Embolisation, curettage + grafting, sclerotherapy, denosumab (selected)

Lesion	Typical patient / site	Key features	Treatment
Giant cell tumour	20-40, **sacrum** (commonest spinal site), vertebral body	Lytic expansile; can metastasise to lung	En bloc or curettage, **denosumab**
Eosinophilic granuloma (LCH)	**Children**	**Vertebra plana** (flattened body, discs preserved)	Usually observation/brace; heals
Osteochondroma	Posterior elements	HME association, cord compression	Excision if symptomatic

3. Primary malignant lesions

- **Multiple myeloma / solitary plasmacytoma:** commonest primary malignant bone tumour; elderly; **"punched-out" lytic lesions**, **bone scan often cold**; **mini-brain sign** (plasmacytoma). **Radiotherapy** (very radiosensitive) + systemic therapy; surgery for instability/neurological compromise.
- **Chordoma:** notochordal remnant; **sacrococcygeal (about 50%) and clival (about 35%)**, mobile spine less; elderly men; **brachyury**-positive, physaliferous cells; **lytic midline destructive lesion with soft-tissue mass**; **en bloc resection with wide margins** (radioresistant to conventional RT; proton/carbon ion therapy adjuvant); high local recurrence.
- **Chondrosarcoma**, **osteosarcoma**, **Ewing sarcoma** (children; vertebral body), **lymphoma**.
- **Enneking staging and the Weinstein-Boriani-Biagini (WBB) surgical staging system** (12 radiating zones and 5 layers) guide **en**

bloc spondylectomy (Tomita) for primary malignant tumours and solitary metastases in selected cases.

4. Metastatic spine disease: assessment

Presentation

- **Pain** (commonest, about 90%): **mechanical pain** (worse with movement/loading: **instability**), **biological/night pain** (tumour inflammation, responds to steroids/radiotherapy), radicular pain.
- **Metastatic spinal cord compression (MSCC):** an **oncological emergency**; about 5-10% of cancer patients; thoracic commonest. Symptoms: back pain, radicular pain, weakness, sensory level, sphincter disturbance (late). **Whole-spine MRI within 24 hours** of suspicion (**NICE**; immediately if neurological symptoms).
- **Steroids: dexamethasone 16 mg** daily (with PPI) as soon as MSCC is suspected (unless lymphoma is suspected and a biopsy is pending).

Decision frameworks

- **NOMS framework (Memorial Sloan Kettering): N**eurological (degree of epidural spinal cord compression: **Bilsky grade**), **O**ncological (**radiosensitivity** of the tumour), **M**echanical (**instability: SINS**), **S**ystemic (fitness, extent of disease, life expectancy).
- **Bilsky ESCC grade (MRI axial T2): 0** bone only; **1a** epidural impingement, no thecal sac deformation; **1b** deformation without cord abutment; **1c** cord abutment without compression; **2** cord compression with CSF visible; **3** cord compression with no CSF visible. **High-grade (2-3)** compression from radioresistant tumour = surgery (if fit).
- **Radiosensitivity (conventional external beam RT): radiosensitive**: **lymphoma, myeloma, seminoma, breast, prostate** (also small cell lung, Ewing); **radioresistant**: **renal cell carcinoma, melanoma, thyroid, sarcoma, GI (colon), NSCLC**. **Stereotactic body**

radiotherapy (SBRT) overcomes radioresistance in selected cases (**separation surgery** then SBRT: decompress to create a 2-3 mm margin around the cord).

- **SINS (Spinal Instability Neoplastic Score, 0-18):**

Component	Scoring
Location	Junctional (occiput-C2, C7-T2, T11-L1, L5-S1) **3**; mobile spine (C3-C6, L2-L4) **2**; semi-rigid (T3-T10) **1**; rigid (S2-S5) 0
Pain	**Mechanical pain 3**; occasional non-mechanical 1; pain-free 0
Bone lesion	**Lytic 2**; mixed 1; blastic 0
Radiographic alignment	**Subluxation/translation 4**; de novo deformity (kyphosis/scoliosis) 2; normal 0
Vertebral body collapse	Over 50% collapse 3; under 50% 2; no collapse but over 50% body involved 1; none 0
Posterolateral involvement (facet, pedicle, costovertebral joint)	Bilateral 3; unilateral 1; none 0

  - **0-6 stable; 7-12 indeterminate (potentially unstable); 13-18 unstable**: **7 or more = surgical consultation**.

- **Prognostic scores: Tokuhashi revised** (general condition, number of extraspinal bone metastases, number of vertebral metastases, visceral metastases, primary site, paralysis: **score 0-8 = under 6 months**, **9-11 = 6 months or more**, **12-15 = 1 year or more**), **Tomita** (primary tumour growth rate, visceral metastases, bone metastases; guides excisional vs palliative surgery), **SORG**, **NESMS**. **Expected survival over about 3 months** is usually required for surgery.
- **Biopsy** when the primary is unknown (and myeloma/lymphoma excluded first: protein electrophoresis, light chains, CT chest-

abdomen-pelvis, PSA, mammography, bone scan/PET).

5. Treatment of metastatic disease

- **Patchell trial (Lancet 2005):** in MSCC from solid (non-radiosensitive) tumours with a single area of compression, **direct decompressive surgery + radiotherapy was better than radiotherapy alone**: **more patients walked (84% vs 57%) and walked longer**, regained ambulation (62% vs 19%), needed fewer steroids/opioids.
  Exclusions: radiosensitive tumours (lymphoma, myeloma, germ cell), paraplegia over 48 hours, multiple compression levels, life expectancy under 3 months.
- **Radiotherapy alone:** radiosensitive tumours, poor surgical candidates, low-grade compression, stable spine; **single fraction 8 Gy** for pain; 20 Gy/5 fractions common for MSCC (SCORAD trial: single 8 Gy non-inferior for ambulatory status in some groups).
- **Surgery: posterior decompression and instrumented stabilisation** (laminectomy alone without fixation causes instability/kyphosis in vertebral body disease: avoid), **separation surgery + SBRT**, **en bloc spondylectomy** for solitary renal/thyroid metastases with long expected survival, **percutaneous screw fixation** (minimally invasive), **vertebroplasty/kyphoplasty** for painful pathological compression fractures without significant canal compromise (**CAFE trial**: kyphoplasty effective for cancer-related compression fractures).
- **Pre-operative embolisation (within 48 hours of surgery)** for **renal cell, thyroid**, hepatocellular, and other hypervascular tumours (and haemangiomas, ABC, GCT).
- **Bone-targeted agents: bisphosphonates (zoledronic acid)** or **denosumab** reduce skeletal-related events (watch **hypocalcaemia, osteonecrosis of the jaw**).
- **Outcome predictors: pre-treatment ambulatory status (strongest)**, speed of deficit onset (rapid onset worse), primary

tumour type.

6. Intradural tumours (brief)

- **Schwannoma** (commonest intradural extramedullary; **dumbbell** through the foramen; T2 bright, target sign), **meningioma** (thoracic, middle-aged women, dural tail, calcification), **ependymoma** (commonest intramedullary in adults; **myxopapillary ependymoma of the filum terminale/conus**; central, haemosiderin cap), **astrocytoma** (commonest intramedullary in children; eccentric, infiltrative), **haemangioblastoma** (von Hippel-Lindau, cyst with enhancing nodule). MRI with gadolinium; surgical excision under neuromonitoring.

Exam pearl: **Metastasis = commonest spinal tumour; thoracic spine; pedicle destruction (winking owl); disc spared. Renal and thyroid metastases are hypervascular: embolise before surgery. Posterior elements: osteoid osteoma, osteoblastoma, ABC; vertebral body: haemangioma (corduroy), myeloma, chordoma (sacrum/clivus).**

Exam pearl: **SINS 0-6 stable, 7-12 indeterminate, 13-18 unstable (7+ = surgical opinion). NOMS: neurological (Bilsky), oncological (radiosensitivity), mechanical (SINS), systemic. Patchell: surgery + RT beats RT alone for MSCC from solid tumours (84% vs 57% walking).**

Exam pearl: **Radiosensitive: lymphoma, myeloma, breast, prostate, seminoma. Radioresistant: renal, melanoma, thyroid, sarcoma, GI.** Dexamethasone 16 mg and whole-spine MRI within 24 hours for suspected MSCC.

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.